

DISCUSSION

SESSION 2

A. R. COOPER,* *Session Chairman*

It is significant that, at this stage of the development of fast breeder reactors, that we are able to hold a full conference on the experiences in running them. The papers in Session 2 cover between them 10–15 reactor years of operation. This is an impressive performance and one basis for the optimistic programmes announced yesterday.

Presentation of Papers

H. O. MONSON—*Paper 2/1*

I would like to bring up to date the paper on EBR-II by describing the period from March 1965, the time at which the reactor power level was first raised to 45 MW, to the present. In March 1965 the integrated thermal power amounted to approximately 800 MW days (Fig. D1); this has risen relatively steadily until today it amounts to approximately 7300 MW days. The integrated electrical power gross is now some 2050 MW days. The 9 week period of no gain shown resulted from a planned outage of the entire plant for general maintenance. Figure D2 shows the periods of power operation and of shutdown. The shutdown periods were required primarily for the fuel surveillance programme and for replacement of spent fuel in the reactor, but there were a few exceptions to this which I will note. First, in respect of the fuel surveillance programme, the limiting burnup of the Mark I fuel element for EBR-II was not established until October 1965. In May 1965 the maximum local burnup reached was approximately 0.8 at. %. In the intervening period, from May to October as each additional increment of burnup occurred, the reactor was shut down and arbitrarily prohibited from starting up again until the fuel element of highest burnup had been removed, cooled, disassembled, and examined, to indicate that an additional burnup increment was warranted. There were four long periods indicated with the number 3 in which the reactor was kept shut down for this reason. The exceptions to this type of shutdown are as follows. In April 1965 we lost almost 2 weeks because of difficulty with the control rod drive grippers; two of the control rod drives had to be removed, repaired, and replaced. Later, in October, we lost about half a week owing to a combination of minor difficulties. In November we lost about 2 weeks because of difficulty with the

* Mr. A. R. Cooper was indisposed.

fuel unloading machine; we had problems of sticking in attempting to withdraw the subassemblies from the primary tank into the fuel unloading machines. Finally, at the end of 1965, in period 7 another 2 weeks were lost while an investigation was made into a hydrogen explosion which occurred in the equipment air lock of the reactor plant. The hydrogen resulted from charging a storage battery inside the equipment air lock. We no longer charge storage batteries inside the air lock. To this total of some 7 weeks in the 14 month period shown can be added a 9 week delay representing the scheduled shutdown for general maintenance.

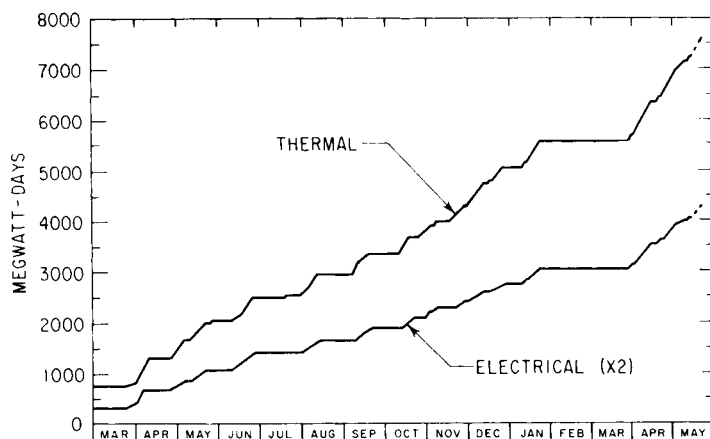


FIG. D1. Integrated power production, 1965-6.

The performance of the reactor in the primary system has been extremely satisfactory; the reactor is highly stable and is readily controlled; the thermal and hydraulic conditions appear to be essentially as predicted. In the secondary system no problems of significance have arisen and we are particularly pleased that the induced activity in the secondary sodium is at a very low level; as an example, the background level in the sodium boiler plant is less than 2 milliröntgens per hour.

With respect to fuel handling, more than 200 core type subassemblies and a large number of blanket subassemblies have been replaced in the reactor since the first approach to power; thirty-seven of these subassemblies have reached the limiting burnup of 1.2% local. Many of the higher burnup subassemblies which had been operating at power densities between 800 and 900 kW per litre of core volume were removed to the fuel cycle facility 9-15 days after shutdown of the reactor. A number of minor difficulties have arisen with the fuel handling equipment but operation of the reactor has not been greatly delayed as a result. Of greater significance is the fact that the time required for fuel handling is short. As examples: in the middle of December, in the period marked 2 (Fig. D2) about one-sixth of the entire

reactor core was replaced in about $1\frac{1}{2}$ days; in subsequent fuel handling periods in recent months more than a quarter of the core in each instance was replaced in less than 3 days; in addition, on several occasions the reactor has been shut down, fuel handling mode entered into, a single subassembly replaced in the core, and the return to reactor operation completed in a matter of hours.

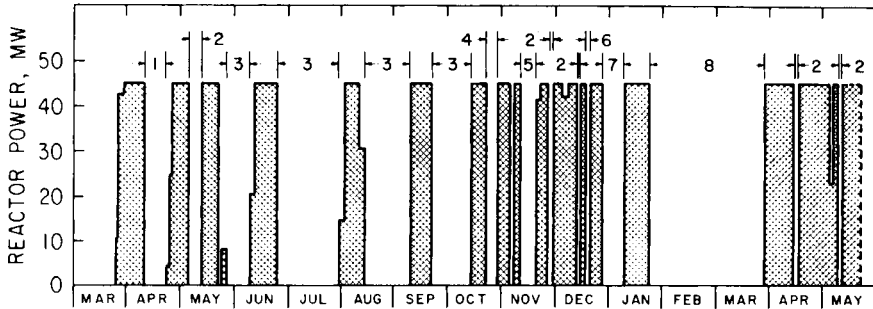


FIG. D2. Power operating history and major events, 1965-6.

1. Fuel handling and control rod gripper repair
2. Fuel handling
3. Fuel handling and wait for fuel examination
4. Fuel handling and minor maintenance
5. Fuel handling (delayed by Fum difficulty)
6. Reactivity experiments with Mark IA fuel
7. Fuel handling and repair airlock damage
8. Fuel handling and scheduled maintenance

The fuel cycle facility operation has been very satisfactory: no major difficulties have occurred although a number of minor ones have arisen requiring revisions to the design of some equipment, some changes in operating procedures, and substantial maintenance. To date the fuel cycle facility has fabricated some 11,000 fuel elements, all by remote control; it has reprocessed 7000 spent fuel elements removed from the reactor and has fabricated about 6000 new fuel elements using that reprocessed fuel. Every fuel element in the reactor at present was fabricated in the fuel cycle facility. About one-half of the present core charge is in its second cycle—that is it has been irradiated once, reprocessed once, and returned to the reactor. A few subassemblies have completed second cycle burnup and been removed from the reactor for reprocessing. It is expected that the first fuel to undergo third cycle burnup in the reactor will be inserted by August or September 1966.

The Mark I or initial fuel element used in EBR-II consists of a pin of uranium fissium alloy inserted in a thin-walled, stainless steel jacket, and it is sodium bonded (Fig. 9 of Paper 2/1). The bottom end of the fuel element is at the right, the top end is at the left. The sodium used for bonding rises some

0.65 in. (1.5 cm) above the top of the fuel alloy pin; there is approximately a 2 in. (5 cm) gas space above the sodium. As burnup accrues and the fuel swells, this displaces some of the bond sodium, and its level rises compressing the gas at the top of the element; the maximum burnup of this type of fuel element is limited by internal gas pressure. At 1.2% burnup, the average fuel element undergoes a change in fuel alloy pin volume of approximately 5%; the poorest performing elements undergo a volume change of the order of 8%. At operating temperature, the resultant pressure inside the fuel element is roughly 20–30 atm. To date, some 15,000 fuel elements have

Reactor zone	Sub-assembly	Numbers of experiments by type					Total in zone
		Oxide fuel	Carbide fuel	Metal fuel	Cermet fuel	Structural material	
Core	XG05	9	3	2	—	5	—
Core	XG06	12	—	2	—	5	—
Core	XG07	—	—	16	—	3	—
Core	XG08	—	8	—	—	11	—
Core	X009	2	6	—	2	9	—
Core	X011	16	—	—	3	—	—
		(39)	(17)	(20)	(5)	(33)	(114)
IB	XG02	1	—	—	—	—	—
IB	XG03	2	—	—	—	—	—
IB	XG04	2	—	—	—	—	—
IB	X010	4	—	—	4	11	—
		(9)			(4)	(11)	(24)
							(138)

FIG. D3. Irradiation experiments in core and inner blanket (as of May 10, 1966).

undergone burnup in EBR-II; of these, about 7000 have been cursorily inspected after burnup in the course of reprocessing; more than 3300 have been carefully inspected in the fuel surveillance programme; not one defect has been found to date. We have made one revision to the Mark I fuel element, and we term this revised element Mark IA. The change consists of shortening the fuel alloy pin by about $\frac{7}{10}$ in. (18 mm) to increase the gas space available for accommodation of sodium displaced due to change of volume in the fuel alloy pin as burnup proceeds. We expect to obtain substantially greater maximum burnup with this new fuel element. At present almost one-half of the core loading consists of the new Mark IA fuel element.

The future plan for EBR-II is to maximize its utilization as a fast flux

irradiation test facility. At present there are 138 irradiation experiments in the core and the inner blanket (see Fig. D3), most of them being in the core. Before the end of 1965 it is hoped to have some 300 irradiation experiments in the core.

K. J. HENRY—*Paper 2/2*

The Dounreay fast reactor has been operating at high power for nearly 4 years and has accumulated as of today 30,000 MW days of operation, about

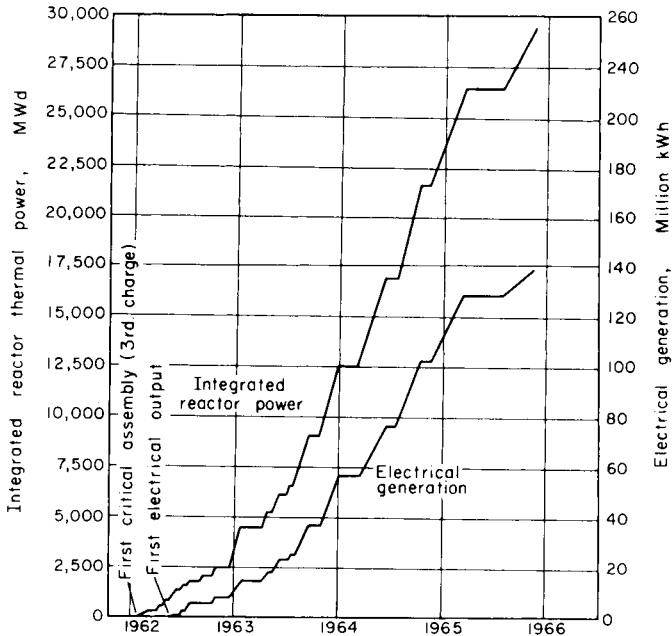


FIG. D4. The operational progress of DFR thermal and electrical power production.

95% of which has been at the full power of 60 MW. It has also generated about 145 million kWh of electricity although I would stress that the prime purpose of the reactor is to serve as a fast flux irradiation test facility. The reactor has amply fulfilled the purpose for which it was built, namely to demonstrate that a high powered, liquid metal cooled fast reactor was a feasible proposition which could be operated safely and easily. The main components of the plant have all performed extremely satisfactorily, and the only major interruption to our scheduled operating programme in recent years occurred a few months ago when some breeder elements could not be unloaded with the standard equipment. Figure D4 shows operational progress up to the present. The shutdown periods last about 3–3½ weeks, during

which we change most of the core and our experimental loading. The rather long period shown in Fig. D4 included a core change but also time spent working on our breeder. The electrical generation is also shown.

The primary and secondary liquid metal circuits have not given rise to any concern about their integrity. The experience we have gained in handling large quantities of liquid metal and in its purification has been encouraging and we feel able to handle the very much larger quantities in PFR. In recent years the level of oxygen in the primary circuits has been kept below about 10 ppm by intermittent periods of cold trapping, assisted possibly by the gettering action of the large surface area of exposed uranium. All the core fuel element uranium is in contact with the coolant. This unusual design, while resulting in an increase in activity in the blanket gas and coolant, has not caused any major difficulties, and there is no evidence of any significant corrosion of uranium; in fact, the uranium and plutonium contents are below the limits of detection, which are about 10^{-4} ppm for uranium and 3×10^{-2} ppm for plutonium.

Our Mark III fuel element, which is now of 7 wt.% molybdenum with 75% enriched uranium, has been tested to an extent which permits us to irradiate the major part of the driver charge on a routine basis to 1.9% burnup heavy atoms and the remainder to 2.2%. This allows operating periods of 80–90 days at full power; we had a run in 1965 of 84 days without power reductions or shutdown. Specimen elements in the reactor—they are typical but have been carefully measured—have reached 3.6% burnup; this limit has been set because at 3.6% the fuel is just touching the can. About twelve such elements have now been irradiated to 3.6% without trouble. We are proceeding rather more cautiously and have not yet called upon the can to place any restraint on the fuel. Examination of this fuel indicates that up to an operating temperature of about 600°C swelling is at the rate of about 10% volume increase per 1% burnup, and from the evidence available it seems that the maximum fuel temperature we normally permit, about 675°C, lies fairly close to a critical temperature for breakaway swelling; above this temperature, there is an increase in the swelling rate. The 675°C fuel temperature is not restrictive as far as operation of the reactor is concerned; the majority of the fuel runs at between 590°C and 620°C, so we still have room for a small increase in rating.

The physics parameters of the reactor have not differed significantly from those forecast. Our knowledge of these parameters is increasing all the time; at each reactor shutdown we spend about 1½–2 days at low power, doing experiments to determine such things as flux distribution and control rod worths. We find that our estimates of temperature in the reactor are becoming more accurate, especially as we are now including in most experimental fuel elements indicators of temperature such as templogs and fusible metal thermometers. On two experimental rigs we have fitted instrumentation

such as thermocouples, but this is difficult; we hope to increase the number of such experiments and are planning to have four in our next operating run. Our knowledge of gamma fluxes is not complete but we are attempting with some new calorimeters to measure these in more detail and so improve our methods of calculation.

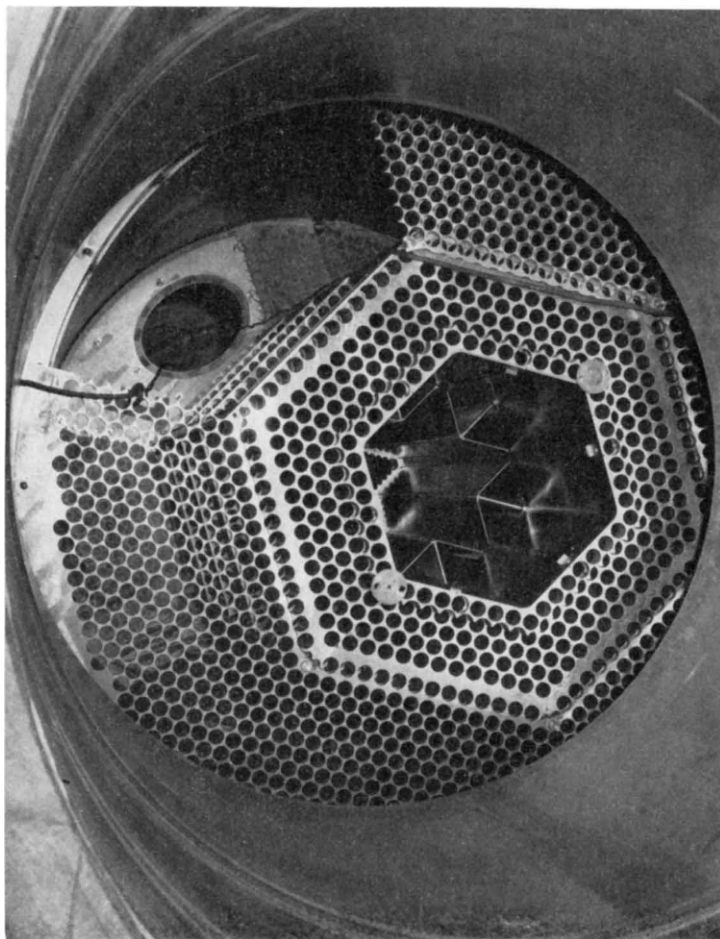


FIG. D5.

The change from a demonstration reactor to a fast flux irradiation facility was effected in the middle of 1963, after the reactor's satisfactory performance at full power was verified. Since then, over 200 different irradiation experiments have been loaded, and about 8% in reactivity is currently being taken up with experimental loading. The extent to which the reactor has

become an irradiation facility can be judged from the following figures: we have had thirty-six mixed plutonium-uranium oxide experiments including

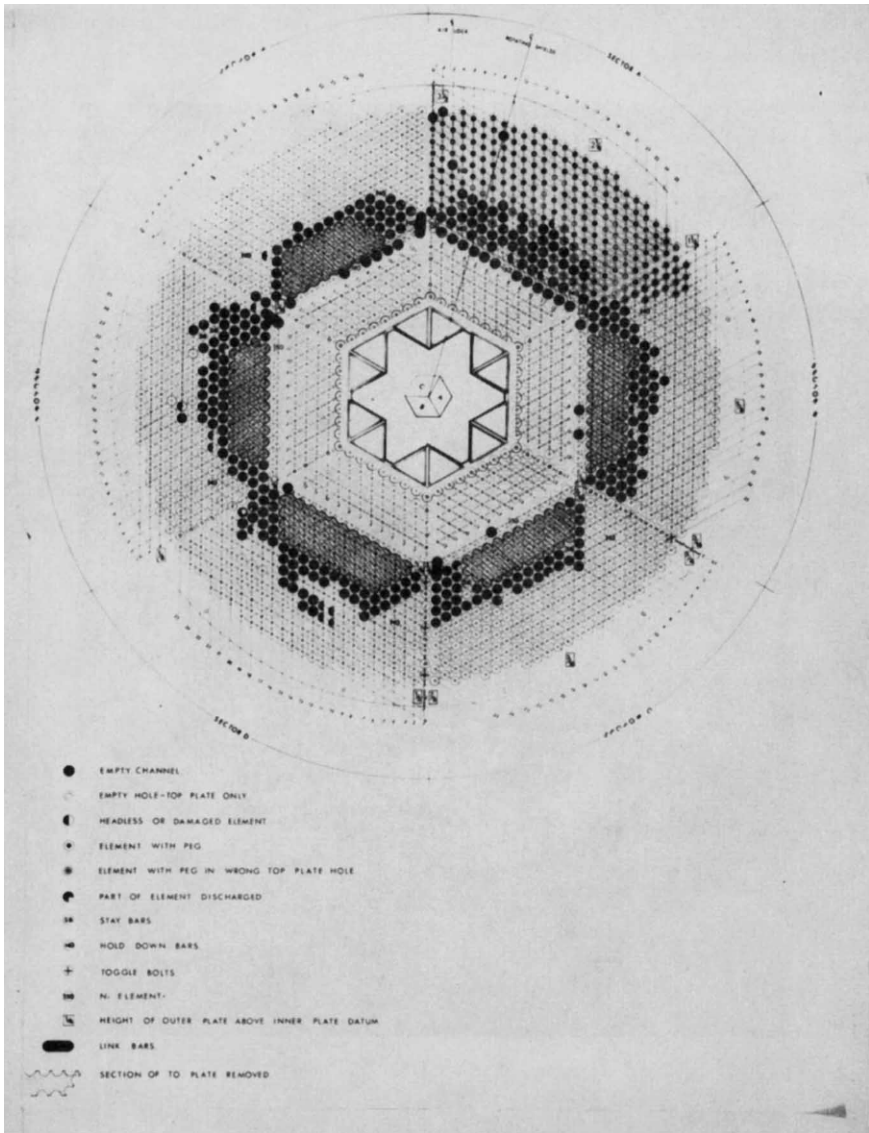


FIG. D6. Plan of outer breeder on completion of modifications.

three subassemblies, six carbide experiments, forty-three cladding experiments and seven graphite experiments. The total number of pins amounts to

248 large plutonium-uranium oxide pins, thirty-six carbides, over 2000 cladding material specimens and 400 graphite specimens. Some of the cladding and all of the graphite experiments are in aid of the thermal reactor programme. It has proved possible during reactor refuelling periods to remove experiments from the reactor, to take pins from their vehicles, to



FIG. D7.

carry out a non-destructive examination and replace them in the reactor before the next start-up. This has required a very tight schedule on the part of our metallurgical division but means that, having succeeded in doing it, very much more information can be obtained from every pin that goes into the reactor. In this way we can get irradiation evidence at each burnup increment up to the maximum.

About a year ago, when the burnup in our outer breeder had reached about 100 MWd per tonne we decided to take some out for examination, and experienced great difficulty in removing them because of swelling. The burnup level was about one-tenth of that at which we would have expected

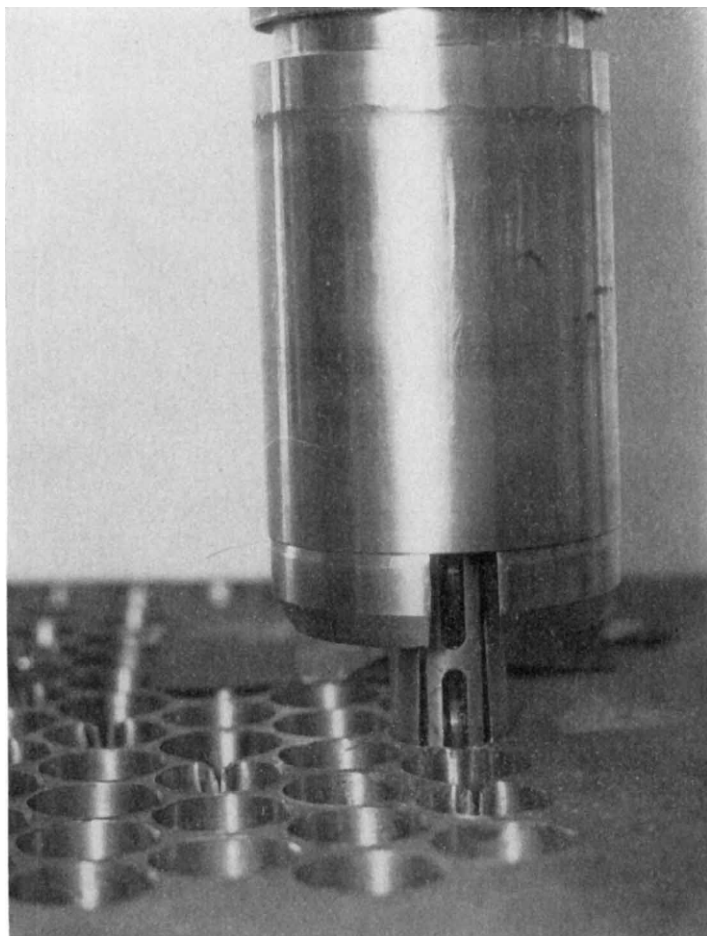


FIG. D8.

to see significant swelling. Figure D5 is a photograph of DFR during construction; it shows one-half of the breeder with the top plate in position. The breeder elements, which are about 9 ft (3m) long, are located at top and bottom in tube plates. We found that in the outer breeder for about five or six rows all the elements were showing signs of swelling; in all, about 600 elements were affected. They had swollen to such an extent that they would not pass through

the top plate. In Fig. D6 the shaded portion shows the areas affected by swelling. We removed about 200 using specially developed heavy lifting gear, but the remainder could not be unloaded with this equipment. We decided to cut away sections of the tube plate, which are the lightly shaded sections shown. This had to be done through a 20 cm hole in the rotating shields, working on top of the active NaK at a distance of about 4 m. Equipment was designed for cutting this plate and removing it, and then for probing into the sodium potassium alloy for pieces of element which had swollen to such an extent that they had been broken in the first attempt at removal. Figure D7 shows a view of the breeder through a shielded lead glass plate. On it is a breeder element standing on top of the tube plate; also shown is a section through a cut plate, and the surface of the liquid metal. Figure D8 shows a piece of plate which we set up in the workshops to test the cutting equipment. The plate had to be cut without leaving any pieces of metal or swarf in the coolant.

These operations took about 3 months and we finally succeeded in removing every damaged element. The cut away regions have been left empty of fuel and the reactor is now operating normally again. The subsequent investigations indicated that because of the low pressure drop across the breeder we were getting some reverse flow resulting in high fuel temperatures and thermal cycling in the range between 500°C and 750°C which led to swelling.

The fact that not only ourselves but also the Russians and the Americans have been able to carry out this sort of operation gives us encouragement for the future, and although further problems will arise we have confidence that they will be overcome.

O. D. KAZACHKOVSKII—*Paper 2/3*

It will be recalled that the reactor uses ceramic fuel and is sodium cooled with an outlet temperature of 500°C. Maximum power is 5 MW, which was reached for the first time in mid-1959. The first loading was entirely plutonium oxide but during the first operating period a partial change to uranium oxide was made. During this period the reactor operated in the range 1–5 MW according to the requirements of the experiment, and accumulated a burnup of 6.5% in the plutonium oxide and 1.4% in the uranium oxide. In the middle of 1965 the reactor was reloaded with uranium monocarbide fuel; meantime a series of investigations was carried out on the discharged fuel.

The first investigations on this discharged fuel began 4 months after the reactor had been shut down; they were concerned with gas release and were carried out in a "hot" laboratory. Gas was blown through each subassembly and directed on to an ionization chamber where its activity was recorded. About 20% of the subassemblies showed activities of three times more than

background, the main active constituent in the gas being krypton 85. The high activity was attributed to pin leakage and it occurred only in subassemblies containing plutonium fuel with more than 5% burnup. Active gas release from these subassemblies increased as their temperature was raised to 600°C—see Fig. 1 of Paper 2/3—and when the temperature was raised to 700°C many of the subassemblies which had previously been leak tight began to release gas too. Investigations into the retention of gas in this fuel showed that 40–45% of the gas is released for burnups of less than 3½% increasing to 50% for burnups above 3½%.

Physical examination revealed two damaged fuel elements (there were nineteen elements in each subassembly). In the stainless steel cladding there were cracks following the grain boundaries of the material and extending over almost the whole length of the element as shown in Fig. 2 of Paper 2/3. There was no plastic deformation of the cladding material. The nature of the damage cannot be explained in terms of gas pressure alone. In the case of undamaged elements, increases of can diameter of about 0.15 mm were recorded, and the cladding material showed increased hardness and reduced ductility. The integrated neutron dose was 2.5×10^{22} nvt.

The technique of discharging irradiated subassemblies into lead was investigated. Ten subassemblies were placed in a bath of lead at 500°C. The lead removed all traces of sodium but there was no noticeable reaction. Gas release was not impeded by the molten lead but stopped completely when the lead was allowed to solidify. Between 1 and 2 curies of activity, mainly caesium 137, remained in the lead bath after immersion of a defective assembly; in the case of lead tight assemblies the leak activity was two orders less.

Radioactive contamination of the coolant was investigated using special coolant by-pass loops in which the flow could be stopped and, if desired, the sodium drained. A scintillation detector in a special collimator was directed at the experimental region of the loop, which allowed analysis to be carried out without removing samples from the circuit. Fission products detected in the circuit were iodine, caesium, barium, lanthanum, zirconium and niobium; the iodine 131 yield was one order less than that of caesium 137. The bulk of the radioactive material migrated to the pipe walls; in fact, once the sodium 24 activity had decayed, draining the liquid metal had very little effect on the general level of radiation from the circuit. The iodine and caesium were removed very effectively by cold trapping; for example, the caesium activity in one trap at shutdown was ten times that of caesium in the rest of the primary circuit, although the trap volume was only one-hundredth of the circuit volume. It is in fact the trap which determines the activity levels for caesium 137 and iodine 131 in the circuit. The zirconium, niobium, barium, and lanthanum activities are much lower and they are not affected noticeably by trapping although these fission products do migrate slowly

to the walls of the circuit. All activity levels were functions of integrated power and of fuel element leakage. Gas blanket activity which was about 0.2 curies per litre was largely due to xenon 133 and xenon 135; iodine 131 activity was negligible, three or four orders less than for xenon. Neon 23 was present due to the sodium np reaction; its total activity at full power was 50 curies, about 5% of the total gas activity.

With monocarbide fuel loading the critical mass is 99 kg, heat rating is 285 kW per litre of core and the peak flux is 4.8×10^{14} neutrons per cm^2 per sec. Temperature and power coefficients are constant and negative, the power coefficient being -8×10^{-4} per megawatt over the range 10–4000 kW, and the temperature coefficient -3×10^{-5} per $^{\circ}\text{C}$ over the inlet temperature range 180–430 $^{\circ}\text{C}$.

W. J. MCCARTHY—*Paper 2/4*

At the present time, the Enrico Fermi power plant is in the middle of a power escalation programme which has been planned to demonstrate proper operation of the plant at 200 MW. As of today, the maximum power level reached is 67 MW. Operation of the reactor and of all the primary system components has been satisfactory in all respects. However, great difficulty has been experienced in stopping small leaks from the water to the sodium side of the steam generators in the secondary system. Since it is desired to repair leaks before damage to adjacent structures can occur, the steam generators have been shut down several times, when the level or rate of change of level of hydrogen concentration in the sodium cover gas has exceeded the preset limits. Various repair techniques are being tried and some understanding has been gained of the behaviour of the hydrogen in the cover gas as a function of sodium temperature.

It is planned to continue the programme of tests, with the achievement of 200 MW, the rated power of the present uranium 10 wt.% molybdenum core, scheduled for July 1966. Several months are to be devoted to tests at 200 MW. Following these tests, modifications will be made to the reactor instrumentation and to the steam and feedwater system, to allow continued operation as a fuel irradiation facility at power levels ranging from 110 MW to 130 MW, depending on the core and test loading. Such operation will be at decreased reactor sodium inlet temperature and increased reactor sodium flow. The combined effects of lower power, lower inlet temperature, and increased flow will be a maximum fuel temperature of about 800 $^{\circ}\text{F}$ (427 $^{\circ}\text{C}$) which is some 300 $^{\circ}\text{F}$ (166 $^{\circ}\text{C}$) lower than the maximum fuel temperature existing under the design 200 MW condition. This reduced fuel temperature is expected to increase fuel life considerably and, depending on the results of a fuel inspection programme which will be carried out during

operation, it is possible that as many as 400 days of operation at the 100–130 MW level can be achieved before the supply of uranium–molybdenum core subassemblies is exhausted. During this phase of operation, up to fifteen special fuel test subassemblies containing a wide variety of fuel and material test specimens will be irradiated in the core. It is hoped that information obtained from the irradiation of these specimens will be of general importance to the fast reactor community, and specifically will enable the fabrication of a mixed plutonium oxide–uranium oxide core for the next loading of the Fermi reactor.

The most recent general survey paper on operations at the Enrico Fermi atomic power plant bore the date March 1, 1965. The low power nuclear test programme, which was well under way on that date, has since been completed. The objectives of this programme were met satisfactorily and complete data were obtained with smaller experimental errors than were anticipated. The results of this programme have been reported in detail and it will suffice to say that the observed physics behaviour of the reactor correlated with the calculated behaviour to a highly satisfactory degree.

The high power nuclear test programme which we are now in the middle of is built around a series of reactivity oscillator tests carried out at 1%, 10%, 50%, and 100% of the designed reactor core temperature rise. We carry out tests at two sets of powers for each temperature rise; for instance, in order to duplicate temperature rise at 100 MW, the reactor is operated at 67 MW and two-thirds primary flow. Figure 1 of Paper 2/4 shows the behaviour of the transfer function at the 67 MW level. The reactivity oscillator in the Enrico Fermi reactor is a rotating poisoned section with a peak to peak amplitude of approximately 5.5 inhours. It has a speed control range from 10 to approximately 0.005 c/s. The zero power transfer function is shown and it may be appropriate to note that we found, to our surprise, that the measured value of the transfer function was some 8% higher than the calculated value. As yet, we have no theoretical explanation for this observation. The transfer function at 67 MW on two loops is a smooth curve agreeing completely with the calculated values. Two such curves are shown, one for manual and the other for automatic control. The meaning of manual control here is that the reactor is brought to power manually, and during the operation of the transfer function reactivity oscillator no control rod movements are made. When measuring the transfer function on automatic control, the automatic control system attempts to compensate for the reactivity variation caused by the oscillator rod, and we observed during earlier power operation that in the range of one-tenth of a cycle per second there was an increase in transfer function due to the operation of the automatic control system. Evaluation indicated that this had no unstable characteristics but we proceeded cautiously and obtained the data shown on Fig. 3 of Paper 2/4. For orientation the zero power transfer function is shown. The next transfer function is at 13 MW

with the reactor on automatic control. Because of the increase above the zero power value, we decided to proceed somewhat more cautiously. We raised power to 30 MW having determined theoretically beforehand that the point of maximum transfer function should go up in frequency and decrease in amplitude, and found indeed that this happened. At 48 MW it is still lower and at 67 MW (which is the same curve as shown on Fig. 1 of Paper 2/4) it is going down quite satisfactorily. As a result of these observations we have completed a reanalysis of the automatic control system and find that this is the way one would expect it to work.

We have carried out power coefficient measurements in detail up to 20 MW, and in less detail up to 67 MW. There are two curves shown on Fig. 5 of Paper 2/4, one for three loop operation and one for two loop operation; one co-ordinate is power and the other is reactivity loss from the start of the operation to the achievement of the power listed on the abscissa. There is no significant hysteresis in the power coefficient either with two loops or three. The data are not really good enough to plot the small change seen here. The power coefficient at 67 MW is approximately as calculated. A preliminary estimate of the time constants of the power override, which is defined here as the product of the power coefficient and power, indicates that the time constants of the various components of the reactivity coefficients are approximately as calculated.

We are carrying out at the present time an extensive series of tests to correlate the thermal power of the reactor with the neutronic observations, and one of the interesting results has been that the intermediate heat exchanger appears somewhat less effective than had been calculated. For a given temperature difference along the primary side of the heat exchanger and a given total temperature across it, less heat is transferred than expected. In view of the present low accuracy of the measurements of flow and temperature, this indication may prove at higher power operation to be untrue.

As indicated, the steam generators have proved the most serious impediment to our proceeding in power operation. The kind of leak we have found has been exceedingly small, of the order of tenths of pounds of water per day. I will briefly review the design of the steam generators and the location of these leaks.

Figure 6 of Paper 2/4 shows an elevation drawing of the Enrico Fermi steam generators. On the left is a picture showing the detailed model of the units which are of shell and tube design. Secondary sodium enters at a design temperature of about 775°F (413°C) through two diametrically opposed nozzles and flows down in the shell over tubes which are carrying water and steam upwards in counterflow to the sodium. There is a single outlet at the bottom of the steam generator vessel which returns the secondary sodium to the pump. Water enters a ring header at the top of the unit, flows down the 1200 tubes in the tube sheet, through a baffled section in the sodium, and

then upwards through convolutions, entering the steam header as superheated steam. At 200 MW the flow is approximately 3 million pounds (1.36 million kg) per hour of sodium, and as indicated here these tubes are relatively small, 0.625 in. (15.9 mm) outside diameter, 0.042 in. (1.07 mm) wall thickness. They are chrome molybdenum alloy tubes.

Figure 7 of Paper 2/4 shows a little detail of the manifolds. The tubes are expanded into the manifolds, which are 5 in. thick on the water side and 8 in. thick on the steam side, and a final seal weld is formed at the top. A number of leaks have been found which have been without exception porosity leaks in the seal welds, mostly in the water manifolds although there have also been many in the steam manifolds.

We have installed instrumentation in the cover gas section of the steam generators capable of detecting hydrogen release. At given hydrogen concentrations in the sodium, varying the temperature of the sodium has a marked effect on the amount of hydrogen which we detect in the cover gas. Often when reactor conditions are being changed, the hydrogen will increase strongly due not to a leak but to hydrogen being released from the sodium. When the limit which we have imposed of about 4000 ppm of hydrogen in the cover gas is exceeded, we feel that we can no longer assume from the instrumentation that a larger leak is not developing. The units are first pressure tested. If there is a pressure loss we search for leaks at the ends of these 1200 tubes, making 2400 individual seal weld examinations necessary. We have found that the quickest technique is to flood the tube sheets with water, to pressurize the shell side to about 100 lb/in² (7 kg/cm²) and look for bubbles. This type of leak is so small that bubble formation often takes many hours. When a definite hole has been found, a repair is made by cutting out and rewelding. We have also used dye penetrant checking to investigate areas which might have a leak but where bubbles cannot be detected. When a dye penetrant indicates a possible flow we have two ways of proceeding, either by rewelding or, recently, by putting a brazing material on the seal welds. This, a high nickel braze material, has been effective. There have been no cases of repaired seal welds subsequently developing another leak. I believe that the units will eventually have all the tubes repaired, either by the welding procedure or by putting a brazed cover on all the seal welds. In the meantime, because of the significance of steam generators to the plant operation, the Detroit Edison Company has asked manufacturers for proposals on replacement tube bundles for these units. The matter of a replacement tube bundle is a very difficult one. The replacement must operate in parallel with the other two. They were designed for about 150 MW of heat apiece and the shells are very full of tubes, so little flexibility is left for the manufacturer of a second unit.

With the exception of the steam generator problem, which must not be minimized, the plant operation has been extremely favourable. In common

with other operators, we find sodium to have its drawbacks but to be a fairly satisfactory coolant. Sodium as a coolant in the United States is not yet accepted, particularly by the utility companies. In the near future the sodium cooled fast reactor business will have to beat by a considerable margin other competing reactors, even different types of fast reactors, in order to overcome this prejudice against sodium as a coolant.

Discussion

R. R. MATTHEWS (Dounreay, U.K.): An important fact to emerge from the Session is that all the fast reactors are now operating at significant power. It is well known that there have been difficulties in getting the reactors up to power but in my view this has been largely due to engineering problems associated with a new coolant technology, and with the experience now available I do not think that these difficulties will arise in future reactors.

There has been no inherently difficult problem common to all the fast reactors. If we go back some 10 years, when designs were first being carried out, there seemed to be a number of serious problems which have since been overcome or have disappeared. Core stability was one: a tremendous amount of analytical work was carried out on this problem and we now find in practice that all the fast reactor cores are extremely stable.

A second point that has emerged is that all the power reactors are now being used as irradiation test facilities, although they were not originally designed for this purpose. I would like to ask the authors of both American papers what plans they have for modifying the EBR-II and Enrico Fermi reactors further to enable them to be used more efficiently as irradiation test facilities. For instance, how will they find the additional reactivity required for fuel experiments? Are they planning to make use of instrumented rigs? What will their philosophy be about taking experimental pins to the failure point, with the possible risk of contaminating the primary circuit coolant? In this connexion I would also like to ask the Russian authors if they can give us any details of the central loop facility which I believe has been installed in BR-5. Can they give us any details of operating experience with this loop? I would also like to ask the American authors what their criterion will be for limiting the burnup of the driver fuel subassemblies, how this will be determined, and what they expect this limit to be.

There is one further point I would like to ask: I noticed that in Paper 2/1 on EBR-II there was a table showing gas blanket activities, and this included the xenon isotopes. I would like to ask if this means that although pins are sealed in EBR-II, fission products are leaking out of these pins, and if this is the case whether the leaks are from pin holes in some of the fuel cans or whether the xenon is diffusing through the can walls so that even with sealed

pins one is bound to have the primary circuit contaminated with fission products.

J. F. W. BISHOP: I infer from Paper 2/3 that the subassemblies referred to in Table 1 were effectively leaking at burnups below 5.1%, and these subassemblies are stated to be in the majority. Is there any indication of the type of leak which is present in these subassemblies?

I then have two questions which I think are important design ones for fast reactor subassemblies. Have the authors found any tendency for pin elongation in their high burnup subassemblies? In particular, is there jamming of the pins in the wrapper or any distension of the wrapper?

Finally, it is seen in the paper that the thickness-to-diameter ratio for the cladding of the carbide fuel is less than that commonly being used in other designs for mixed oxide fuel. Do the authors intend to conduct experiments with greater thickness-to-diameter ratios in the light of recent observations of fuel swelling?

S. M. BLAGOVOLIN (U.S.S.R.): I wish to ask some questions about the steam generator on the Enrico Fermi reactor. I am interested in the number of tube failures which occurred, and in the method of repairing them. Did leaks occur as a result of faults in manufacture or were they due to high stress under operating conditions?

Y. N. KOSKIN (U.S.S.R.): Dr. Monson said in his presentation that difficulties were encountered with control rod mechanisms and fuel handling mechanisms in the reactor. Could I have more details of this problem? Were the failures due to the presence of oxides or were there other factors? Did the failures occur below the sodium level or in the gas spaces?

W. MARTIN (Oakridge National Laboratory, U.S.A.): In Paper 2/3, concerning the cladding material which had failed—in this 18/9 titanium steel, what are the titanium, and nitrogen and carbon contents? What was the grain size of the material prior to irradiation? Since the elongation of this steel on tensile test was only $\frac{1}{2}\%$ after 5×10^{22} nvt, what steps are being taken to improve the ductility of the tubing?

J. ALLEN (U.K.): In Paper 2/4, I was interested to hear that no leaks had been observed in the tubes of the steam generator in the Enrico Fermi reactor. Would a different response have been expected in the blanket gas if the leaks had been beneath the sodium? Secondly, since no leaks have been seen in the single wall tubes, can we infer that the new steam generators under consideration will be of single wall tube type? Thirdly, would the author

propose to use the same type of weld geometry as currently used on the Enrico Fermi steam generators?

A. G. FRAME (UKAEA): In Paper 2/1 it is suggested that the sticking of the rotating shield may have been due to deposition of sodium vapour. Has there been any other sign of sodium deposition particularly in pumps and control rods? Secondly, the steam generator in EBR-II has double tube plates. Has there been any indication of weld failure at this tube plate? Thirdly, have any investigations been carried out into vibration problems in the primary circuit of EBR-II?

In Paper 2/3 an interesting reference is made to the use of lead for decontaminating subassemblies. Does the author anticipate using lead as a standard method of decontaminating subassemblies?

A. G. EDWARDS (UKAEA): I would like to ask the members of the American and Russian delegations a general question concerning the use of an oscillator. The situation at present appears to be that at Dounreay we have a rotating oscillator of rather small reactivity worth; it does not function with great mechanical reliability, and we have been prepared to operate for periods without it. From Paper 2/1, I think we can infer that their last power increment was made after their oscillator had broken, and it appears that they also are prepared to rely upon other methods of establishing stability. I am not sure of the Russian position and I would like to address this general question: Is an oscillator a key piece of plant equipment or is it an optional extra which can be dispensed with?

J. L. RATIER (CEN, France): In Paper 2/3, have the Russians seen any evidence of chemical reaction between oxide fuel and sodium, either from examination of fuel pins after irradiation or from taking samples in the sodium circuits? Secondly, in what chemical form are the fission products which have been noticed in the coolant?

J. P. MUSTELIER (CEA, France): Concerning fuel elements in BR-5 in Paper 2/3, have the authors seen an increase in the cladding diameter of the gas plenum region of those elements in which fuel swelling occurred? Secondly, I understand that the irradiated fuel pin develops a central void which increases with burnup. Would the authors confirm this and comment on the fact that the central void does not play an important role in preventing deformation of cladding during swelling? I would also like to know the maximum specific power of the uranium carbide fuel in the new BR-5 core.

J. L. PHILLIPS (Dounreay, U.K.): As an operator I found Paper 2/3 particularly interesting because of the implications of the possible use of

vented fuel elements on maintenance operations. Professor Kazachkovskii described a certain amount of maintenance carried out in the primary circuit following a period of operation during which there had been a major increase in fission products in the circuit, and he says that cold traps were used to purge it of radio active contamination. I would like some indication of how effective this purging operation was, and how much plating of fission products and alpha activity remained on the pipes and equipment. Perhaps rather than give me numbers on this he might indicate on the basis of his experience how feasible he would consider the proposed operation of maintenance on, say, a hydrostatic pump bearing. Secondly, has this experience encouraged the proponents of the philosophy of the use of vented fuels for say BN350?

G. DENIELOU (CEA, France): We have been listening to most interesting statements about high temperature reactors. But the claims of another strong contender, heavy water, has not yet been heard here. I do not wish to be suspected of having connexions with heavy water people, but it would be as well to recall a recent statement by Dr. Lewis of Canada, who said that it was necessary for a fast reactor to achieve an n_{eff} of 5×10^{23} in the cladding in order to be economic and that from simple consideration of n , α reactions this was clearly impossible. Is there any comment from someone in charge of a fast reactor about this statement?

C. P. ZALESKI (France): I want to ask Dr. McCarthy whether on the Enrico Fermi reactor any recent measurements of the effectiveness of the shielding have been made, and if so whether the design predictions were realized or not.

Authors' Replies

H. O. MONSON—*Paper 2/1*: As far as the changes planned for EBR-II to improve its performance as an irradiation test facility are concerned, I described one change, the shortening of the fuel alloy pin, in going from Mark I to Mark IA fuel elements. In fact we made a second change; the enrichment of the fuel alloy was altered from 48.4% to 52.5%. This increased the reactivity and almost doubled the test volume available and, as a result, the number of experimental subassemblies which can be accommodated has risen from about eight to fifteen. We are looking for additional improvements in the driver fuel. Further enrichment will probably be made to accommodate more experimental subassemblies, but more important, we would like to raise the plant factor by increasing the maximum achievable burnup. There are a number of routes which can be followed. We have already, as indicated earlier, enlarged the expansion space available but we can go further. Only minor changes were made initially in order to avoid modifying the fuel

fabrication equipment. With relatively minor alterations to the fuel cycle equipment we can make changes in the length of the fuel element jacket, and by this means substantially increase the volume available for accommodation of the sodium which is displaced by fuel alloy growth. There are other simple modifications, such as a slight reduction in the diameter of the fuel alloy pins inside the jacket. We are looking hopefully at other alloys for greater resistance to swelling, so we think that in the near future we can make considerable improvement in maximum achievable burnup.

We are also considering changing the inner blanket and perhaps part of the outer blanket from uranium to stainless steel, nickel or some other material; this will flatten the flux somewhat and increase reactivity slightly. Additionally, we are attempting to develop new types of irradiation subassemblies (vehicles for insertion of experiments in the reactor) which will afford higher coolant temperatures in the test sections. At the moment we are limited to a maximum coolant temperature of about 1000°F (540°C). We hope to raise this to close on 1200°F (650°C).

The criterion for burnup is a little nebulous at this stage. There was no need to attempt to get the last bit of burnup from Mark I elements, of which we had a limited number and of which there are now only a few left in the reactor. With the Mark IA, the limit will be based upon maximum pressure in the worst placed fuel elements. We would like to be conservative in this respect, at least for some time to come. It is our intention in the relatively near future to run some control tests which will produce failure in a small number of fuel elements.

The question was raised about primary cover gas activity. We believe that the activity seen, which is of low level, does not come from leakage in fuel elements. This opinion is based on the examinations we have made of a large number of elements, none of which has shown any leakage. We think the activity is from tramp uranium in the sodium. The concentration of uranium in the sodium may approach a few hundredths of one part per million, but in 320 tons of sodium this amounts to quite a lot of uranium. The system activity remains at such a low level that, although during power operation at 45 MW the sodium 24 activity amounted to about six-tenths of 1 curie per litre of sodium, when during the 9 week shutdown in February and March 1966 we lowered a monitor into the primary tank after the reactor had been down for several weeks and sodium 24 activity had decayed, we found that just above the surface of the 300 tons of sodium the activity was less than 15 mr per hour.

I was asked about the failure of some of the fuel handling mechanisms in EBR-II. One such failure occurred with control rod drive mechanisms. The difficulty was with the gripper device at the end of the control rod drive shaft which uses two jaws for grasping the top ends of the control rods. The jaws are cam actuated and trouble arose because the cams were not securely

mounted within the gripper. One cam broke off, stayed in the gripper, and was recovered when the control rod drive shaft was removed. We have taken out two control rod drives and repaired them.

There has also been some difficulty with the final unloading machine. Spent fuel subassemblies are removed from the reactor, or from the storage basket, vertically up through the transfer tube, through the transfer port and into the unloading machine. We found that the gripper associated with the fuel unloading machine which carries the subassemblies upwards retained a fair amount of sodium; it failed to drain properly. After the subassembly was raised into the fuel unloading machine, some of the sodium from the gripper continued to drip; it collected at the top end of the fuel transfer port, so when attempts were made later to remove additional subassemblies, the solidified sodium caused trouble. This problem was solved by redesigning the gripper to facilitate draining of the sodium.

Thirdly, the transfer arm has given some difficulty as mentioned in the paper. That difficulty worsened, so, during the prolonged scheduled shutdown in February and March 1966, we removed the transfer arm, repaired and reinstalled it. It may be of interest to you that during preoperational testing of EBR-II the two primary pumps failed. The pumps, which weighed roughly 20 tons each, were lifted into a $6\frac{1}{2}$ ft (2 m) diameter caisson some 30 ft (9 m) in height, using a piston inside the cylinder to form a gas tight seal. We hoped to use a similar technique for the transfer arm but found that space was not available for this caisson and instead we used a neoprene/nylon bag. All the preparations for this removal together with the actual removal, the repair of the transfer arm, and its reinstallation were accomplished in approximately 3 weeks. The transfer arm difficulty was purely mechanical; it had nothing to do with the sodium environment.

Mr. Frame asked about the deposition of sodium in the gas regions and in components removed from EBR-II. This varied somewhat from component to component, but generally we have found significantly less sodium oxide or condensed sodium than might have been expected. During the prolonged shutdown 2 months ago, when the transfer arm was removed and we had a very large opening in the primary tank, we inserted some viewing equipment in it. With this we could look up into the annular gap between the rotating plugs and the fixed structure. Only small amounts of sodium were seen. However, visibility was not what we would have liked, and this evidence is not conclusive; we still do not know for certain what causes the sticking of the rotating plugs but the difficulty has not worsened.

O. D. KAZACHKOVSKII—*Paper 2/3*: The first question asked by Mr. Matthews was about the central loop of reactor BR-5. It is difficult to make a loop with an independent cooling channel, especially in a reactor with a small core; the diameter of our core is only 28 cm. In practice it was impossible

to have access to the channel from the top because there is so little space there, so we decided to get the fuel element in from underneath. The channel itself has a diameter of 25 mm. The difficulty was that gas accumulated at the top of the channel and the circulation of the coolant became restricted. When the pump flow was reduced the pressure in the channel decreased; as a result the gas expanded and circulation in the channel stopped altogether. After these experiences I think that a loop in a reactor of this type is unlikely to be very useful. Of course certain interesting results were obtained from the experiment. Much more information will be acquired when we put samples inside the active core instead of in separate subassemblies in the reflector.

Dr. Bishop asked how leakage of gas from fuel elements takes place. I showed on the screen the shape of the cracks which appeared on the fuel elements; the gas escaped through these cracks. There was no sign of elongation and no evidence of jamming of pins in subassemblies. Dr. Bishop also asked about diameter/thickness ratio. The core of the reactor BR-5 is filled with fuel elements of uniform design and we are not going to change now. We intend to burn them up until we find some damage to the elements. At the same time, we will carry out investigations on these fuel elements in thermal reactor loops.

I was asked about the composition of the steel used in our reactors. It was a standard steel with a nominal composition of titanium 1% and carbon less than 0.1%. Its elongation on tensile test in the unirradiated condition was in the range 30–40%. We expect to be able to improve on this steel by introducing nickel.

There was a question concerning lead: we are going to investigate the technique of discharging fuel elements into lead on a larger scale in BN-350. There were also many questions which touched on the behaviour of radioactive materials in the sodium. This matter has not yet been studied properly and, while it is plain that cold traps remove some of these products, we do not always get consistent results and are going to conduct more extensive tests on BN-350.

Questions were asked about expansion of the fuel element. Voids do not always develop in the centre of the element; indeed, in most cases, there is no sign of a central hole on cutting open the element. There may be a relationship between the central void and the stress in the cladding.

W. J. MCCARTHY—*Paper 2/4*: As the Chairman suggested, these reactors were never designed as fuel test facilities. In fact, during the design of the Fermi plant, many attractive features which might have helped in this respect were more or less ruthlessly suppressed. The Fermi plant, for instance, cannot be adapted in any practical way that we can see to monitoring in detail any test subassembly operations. We are designing a standard test

subassembly to be used for all fuel tests. It will consist basically of a fuel subassembly which will fit into our lattice; it will have four thimbles instead of our normal 140 pins, designed in such a way that it can be made independent of the temperature of the sodium by environmental insulation. The temperature limit in these thimbles will be 1200°F (650°C) which includes hot channel factors. We will need a number of different kinds of instrumentation in the plant to handle plutonium, which we are not yet equipped for, but basically very little will be or can be done to improve it as an irradiation facility.

There is one possible exception. There are at present eight safety rod positions in the core; we operate with seven safety rods, the eighth position being used for the oscillator rod. It is possible, we believe, to operate another safety rod position as a material irradiation hole in the reactor. This would be in many respects a very fine irradiation position; it is in the direct centre of the core. There are two possible ways of doing it: first, by using the safety rod as a carrier for material specimens, replacing the poison sections of the rod itself with material specimens and then handling it through normal fuel handling procedures; a second and extremely expensive way, but possibly with very satisfactory results, would be to use this hole down to the centre of the core as an instrumented irradiation "glory hole" as it has been called. At present estimates are in excess of one million dollars for designing and making the necessary modifications for this test position, but the Atomic Energy Commission has requested us to begin looking at the possibility of such a scheme.

As for reactivity, the core section of the Fermi reactor is relatively flexible. We are operating now with 105 subassemblies in the core. It is possible to locate 139 subassemblies in the section of the reactor fed by the primary coolant pumps. The criterion we have set for irradiation specimens is that a test subassembly containing fuel irradiations when placed in the reactor core must, together with the fuel subassembly which it has displaced to the edge of the core, preserve the same reactivity as before. Our failure-of-fuel criterion is necessarily a very strict one. We are going to do everything possible before accepting subassemblies and fuel irradiation specimens to ensure that they will not fail.

The criterion for limiting burnup of driver subassemblies will essentially be based on visual hot cell examination of fuel subassemblies removed at intervals of approximately two-tenths of 1% burnup. It is hoped that the present molybdenum fuel can reach 1.8 at.%. Information from Dounreay suggests that we may be able to exceed this, but our criterion will be based on examination of our own irradiated fuel subassemblies.

I was asked by Dr. Blagovdin how many genuine leaks—those found by seeing bubbles—we have found. There were about 100 such leaks and they have all been repaired by rewelding with the Cr/Mo alloy welding rod. His second question concerned the supposed cause of the leaks: is it the welding

procedure or stress in the units? It appears to be the welding procedure. I have a slight caveat to make, however, namely that there have been some cracks immediately adjacent to tubes in the water manifold on two of the units. These may be caused by thermal-shock-induced stresses due to changing temperatures following reversal of flow at very low powers in these tubes. We are repairing these by rewelding. Stress analysis of the main tube sheet has been repeated and has shown that the ligaments themselves are not highly stressed.

Mr. Allen requested information on the possibility of a different response by the hydrogen instrumentation to a leak below the sodium compared with one above. About 2 years ago we did tests on steam generators in which we injected water at varying rates and levels under the sodium and also into the gas space above the sodium. My recollection of the results is that for small leaks of the order of 0.10 lb/in^2 (0.0007 kg/cm^2) per hour there was no significantly different response between those in the sodium and those above it. For medium leaks, the injections of water below the sodium were noticed some time later than similar injections into the gas space. For large leaks there was again no time delay.

I was asked my opinion of single walled tubes. For a number of obvious practical reasons the replacement units, if such are bought for the Fermi steam generators, will almost certainly be single walled tubes. It is important to remember that the choice of single walled tubes for steam generators at Fermi was not made without consideration of the cost of double walled tubes. My recollection is that double walled tubes would have been about twice as expensive, perhaps three times, for units of this capacity. The general feeling we have is that single walled tubes are still very much favoured. In future designs a lot of attention must be given to the vibration problem which caused our single walled tubes to fail 2 or 3 years ago. We are confident that the problem of sealing the tube end welds can be coped with.

Mr. Allen's third question had to do with the weld geometry. At present the tube to tube sheet welds are not radiographable. The weld is designed as a seal weld only; the roll was intended to provide the strength, and the weld is made between the Cr/Mo tube and a forging—which is not the current best practice. I believe that in a new unit an attempt would be made to have radiographable tube welds; they would probably be designed as strength welds rather than specifically as seal welds. The suppliers to whom we have spoken would probably try to overlay tube material on to the tube sheet forging so that the weld could be made between similar materials.

Mr. Edwards spoke about oscillators. I think it is important that we have techniques for measuring the kinetic response of reactor cores for a long time until we understand burnup effects better, and also understand the detailed kinetics of the different kinds of core being considered for fast

reactors. The oscillator is not the only way of doing this. Our oscillator has worked well up to 67 MW, but I agree with Dr. Edwards that it is undesirable to have any rotating piece of machinery inserted at the end of a 30 ft (9 m) shaft into a sodium atmosphere and a flux of 2×10^{15} neutrons/cm² sec. We have investigated means of determining kinetic response by control rod ramps, by power noise analysis and similar techniques, and I suspect that we will attempt to dispense with oscillators in future.

The last question was from Mr. Zaleski. We conducted detailed shielding investigations on the Fermi reactor at 20 MW and found no primary shield deficiencies at all. We have also done a check at 67 MW, and that conclusion remains firm.

Written Contribution

D. HAGGER (The Plessey Co. Ltd.): Two types of coolant have been proposed for fast reactors, liquid metals and high pressure gases. In liquid metal cooled reactors the coolant often becomes highly radioactive with a fairly long half-life. In gas cooled reactors the coolant does not become as radioactive nor is the half-life as long.

Two general types of fuel element have been considered, vented and non-vented. In vented fuel gaseous fission products are allowed to leak into the coolant often after an appreciable hold up (5 days as quoted in Paper 2/3). In non-vented fuel the gaseous fission products should be retained by the element cladding.

Obviously only non-vented fuel would be used in gas cooled fast reactors. In liquid metal cooled reactors either could be used from a radioactivity point of view since the gaseous fission product activity will be a small fraction of the circuit activity and cleanup systems have been developed (Paper 2/3).

The choice of BCD systems will depend mainly on the type of coolant. In gas cooled reactors the type of failure which is to be detected will probably be of the same type as that in gas cooled thermal reactors. The solution will probably be best solved by using a precipitator detecting gaseous fission products.

In liquid metal cooled reactors, the type of failure most feared is that which would arise if a pin burst releasing a puff of gas big enough to blanket further pins. This could create a rise in temperature high enough to burst these pins releasing more gas and propagating the failure to catastrophic proportions. Such a burst would have a very short time constant, much shorter than the time constants of proposed BCD systems. If, however, vented fuel were used the large rise in pressure could not arise in the first place.

To summarize, in gas cooled fast reactors a BCD system based on the precipitator can be used to detect failures in non-vented fuel. In liquid

metal cooled fast reactors the use of vented fuel elements removes the need of a BCD system, other than visual inspection at reactor shutdown, and BCD systems with time constants short enough to deal with the most important type of failure of non-vented fuel have not and probably cannot be designed.